

# An Intelligent System for Diagnosing Thyroid Disease in Pregnant Ladies through Artificial Neural Network

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## ABSTRACT

In the modern era, the field of medicine is witnessing an enormous growth in methods of diagnosis to the methods of treatment and the abundance of information which is not reaching the front lines in an appropriate time deters the ability to correctly choose an optimal line of treatment. The explosion of the availability of information about the methods of diagnosis and various treatment options requires some sound computational framework which enables to optimally use the patient-specific diagnosis as well as a suitable treatment, and one such computational framework is artificial intelligence (deep learning), which performs the tasks with the help of computers, normally requiring human intelligence such as image processing, speech recognition, audio and/or visual perception, translation, and decision making. The recent development of Artificial Intelligence has provided such a platform to handle, compile, analyze, and interpret vast data in order to enable doctors to shift the focus from diagnosis and identification of patient-specific treatment to actual patient care. In this paper, we present an effective and efficient diagnosis system using Artificial Neural Network (ANN) for Thyroid Disease (TD) diagnosis by first training the model based on training data set and then use the model to predict the presence of disease. We have compared the prediction results of three different methods i.e. Multiple Regression, Random Forest (Ensemble Learning Method for Classification), and Artificial Neural Network (Deep Learning). When the accuracy results are compared for the prediction results, ANN delivers the highest accuracy as compared to the other two methods. The accuracy predictions for the Test Dataset of ANN, random forest and Multiple Regression are 98.22%, 22.46%, and 2.42% respectively. Given the thorough design and problem formulation, powerful computational frameworks such as an artificial neural network can help problem specific decision making as it yields very accurate prediction results as compared to other traditional models. Future scope as described in this paper, outlines the potential research area and the possibility of integration of artificial neural network for providing patient-specific personalized diagnosis and feasible and optimal treatment options which will minimize the risk and the time of recovery.

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## 1. Introduction

Nowadays, Thyroid is one of the most prevalent public health problems, especially in pregnant ladies. It impairs metabolism, mental functions, energy level, weakness and bowel movements. During pregnancy, almost 70% of all diagnosed with thyroid disease are due to deficiency or excess of iodine in the diet. According to the Indian Society of Thyroid, the women in developing countries are pregnant for as much as half of their productive lives and therefore are at an increased risk of thyroid during this time. Globally, the thyroid has become the utmost hurdle in pregnancy. Decision Making in Medical Sciences has undergone a lot of improvements to predict the exact output of any disease. It is also becoming a very essential part of today's world because of the awareness of the various health problems. A group of causing factors of a particular disease or problem is given as inputs and based on the given input values, a decision is arrived at. The decision making can be done by using an artificial neural network. In this research paper thyroid problems (TSH levels, iodine deficiency, Hypothyroidism) in pregnant ladies are diagnosed using an artificial neural network approach. The efficiency of the network is analyzed. Artificial Neural Network for medical parameters such as TSH levels T3 and T4 and thyroid deficiency in pregnant ladies have been considered. The artificial neural network has been trained with the help of various parameters by using the existing data (training dataset), and then the occurrence of thyroid disease is predicted for comparison with the existing data to measure the accuracy of prediction.

### 1.1 What is Thyroid Disease?

The thyroid is a butterfly-shaped gland found just below Adam's apple of our neck. It is responsible for the metabolism activities of our body. When it functions properly it produces two hormones called Triiodothyronine (T3) and thyroxine (T4). A hormone called Thyroid Stimulating Hormone (TSH) which is secreted by the pituitary gland is also responsible for T3 and T4 hormones. TSH, T3, and T4 decide the health of a person.

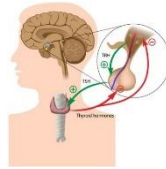


Figure 1. Pictorial representation of the thyroid gland (Bridget Brady, 2017)

Overactivity of thyroid hormones results in hyperthyroidism whereas, under-activity of the same results in hypothyroidism. Thyroid disease can be diagnosed using the following tests (i) Radioimmunoassay test for TSH, T3, and T4. (ii) Scanning for goiter or nodules etc. Thyroid disease is found in KERALA based people because of the iodine content problem (Unnikrishnan & Menon, 2011). People in the age group of 20 to 40 are mostly affected by thyroid disorders. Women are having more risk than men in developing thyroid disorders. They can develop thyroid disorders during their pregnancy also (Vanderpump et al., 1995).

## 2. Related Work

(Nazari Kousarrizi, Seiti, & Teshnehlav, 2012) proposed feature selection and classification approach on two datasets for thyroid disease diagnosis. The study exhibited the accuracy of 98.62% of the proposed system which is the highest accuracy reached so far. Bastias et al. (2011) proposed a model to assist in the diagnosis of malignancy of thyroid nodules based on neural networks. The researchers had taken 1097 patients data to identify thyroid nodules. The study proposed that the health care field will benefit from the incorporation of leaning capabilities in decision system models. Sarasvathi and Santhakumaran (2011) investigate the diagnosis of thyroid disease through the artificial neural network with backpropagation algorithm. The proposed approach had been trained and tested using 80 samples. The study reveals that the backpropagation algorithm was proved to be having high sensitivity and specificity. (Ozyilmaz & Yildirim, 2002) Investigates the diagnosis of thyroid disease using different artificial neural network methods, which were Multilayer Perceptron's, Radial Basis Function, and Conic Section Function Neural Network. The results exhibited that feedforward neural network successfully detects thyroid disease. (Margret, 2012) Aims to diagnose the thyroid disorders using decision tree attribute splitting rules which were Information Gain, Gain ratio, Gini Index, Likelihood ratio, Chi-Squared Statistics, and Distance measure. It is very clear from the results that normalized based splitting rules have high accuracy and sensitivity. Chen et al. (2013) utilizing Fuzzy K-nearest neighbor (FKNN) and presented an effective and efficient diagnosis of Parkinson's disease (PD) diagnosis. The researchers have compared the proposed system with support vector machines (SVM) based approaches. The experiments and results prove that the proposed system performs significantly and achieved an accuracy of 96.07%. Pandey, Gour, and Sharma (2015) proposed a robust classifier (Ensemble of C4.5 and Random Forest) for the accurate prediction of thyroid disease. They have compared the proposed system with other classifiers also and the results show that the proposed system achieved an accuracy of 99.47%.

Sehgal and Taneja (2015) presented a systematic approach for earlier diagnosis of thyroid disease using a back propagation method. The experiments and results prove that the Levenberg Marquardt Method had better performance as compared to a simple gradient descent algorithm. (Anusha & Sathiaselalan, 2016) tried to improve the prediction accuracy of diagnosing diabetes disease with the application of evolution clustering multi-objective optimization algorithm (ECMO). The experiments and results prove that ECMO achieved maximum accuracy for the dataset. For the diagnosis of thyroid disorders, Ahmad et al. (2018) presented a unique intelligent hybrid decision support system by using Linear Discriminant Analysis (LDA), Adaptive Neuro-Fuzzy Inference System (ANFIS), and k Nearest-Neighbor (KNN) Weighted Preprocessing. The proposed system achieved an accuracy of 98.5%. (Sharma, 2017) did a literature review to find out the suitability and appropriateness of a neural network for disease prediction. The researcher had taken 100 significant contributions from 1991 to 2016. The study revealed that different various architectures of the artificial neural network (ANN) were found suitable and appropriate. (Sidiq & Khan, 2017) presented an overview and comparison of existing data mining techniques used for diagnosing thyroid diseases. It is found that the neural network performs better than other techniques.

(Kumar & Abhishek, 2012) investigates the diagnosis of kidney stone disease using three different neural networks, which were Learning Vector Quantization (LVQ), Two Layers Feedforward Perceptron trained with Backpropagation Training Algorithm and Radial Basis Function (RBF). The

researchers had compared all three neural networks with each other. The experiments and results prove that multilayer perceptron with two hidden layers and the Backpropagation Algorithm was the best model for diagnoses and achieved an accuracy of 92%. (Kaur, 2013) presented a comparison between two artificial neural network algorithms, Multilayer Back Propagation (BPN) – a supervised approach and Self Organizing Maps (SOM) – an unsupervised approach for the diagnosis of thyroid disease. The results proved that the unsupervised Self Organizing Maps (SOM) network performed better and achieved higher accuracy. Zhang and Berardi (1998) discussed the investigation of Neural Network in Thyroid Function diagnosis by ANN. Guoqiang Zhang and Eddy Patuwo (1998) presents a state of art survey of artificial neural network (ANN) applications in forecasting. The researchers have explained about the artificial neural network (ANN), Applications of ANN as forecasting tools, different architectures of ANN, unique characteristics of ANN, relative performances of ANN in forecasting and issues in ANN modeling in detail.

Kodaz, Işcan, and Babaoğlu (2009) proposed a method for diagnosing thyroid disease with Artificial Immune Recognition System (AIRS). The proposed method achieved an accuracy of 94.82% using 10-fold cross-validation (CV) method. The researchers stated that the proposed method would be helpful in diagnosing thyroid function. Saleh, Jabr, and Abdulhussein (2016) proposed a hybrid intelligent system, which was a combination of Neural Network and Fuzzy Logic for the diagnosis of thyroid disease. The experiments and results showed that the proposed system accuracy was 100% for training, and in the range of 87% to 95% for testing. The researchers stated that the proposed system could be used in the diagnosis of thyroid disease. Amato et al. (2013) explained that artificial Neural Network was a powerful tool to help physicians to perform and solve medical diagnosis problem. Sundaram and Renupriya (2016) had investigated the artificial neural network (ANN) accuracy for diagnosing thyroid disease. The researchers proposed that feedforward networks and Elman Networks were used for modeling nonlinear dynamic systems like thyroid. The proposed method achieved an accuracy higher than 90%. Nagar (2015), for prediction of thyroid disease, compared two neural network models namely, Multi-Layer Perceptron (MLP) and Radial Basis Function (RBF). The MLP had been trained and tested with the backpropagation algorithm whereas the RBF network was trained and tested with SPSS software. The experiments and analysis proved that the radial basis function network can be successfully used for diagnosis of thyroid disease. Ammulu (2017) proposed a random forest approach to predict hypothyroid disorder. The experiment and results provide improved accuracy, precision, recall, and F-measure by comparing the proposed method with the Linear Discriminant Analysis (LDA) algorithm. (Parveen, Nabi, Memon, Zaman, & Ali, 2016) presented techniques of Artificial Neural Network which comprehensively enables the medical science to predict the occurrence of diseases. The study revealed that for improving the diagnosis of the disease, improvements in the drug prescription development and proper maintenance of recordkeeping (patient's history) is also essential. (Ahmad et al., 2018).

Rastogi and Bhalla (2014) proposed a feed-forward neural network and displayed how it can be used to classify the thyroid disease using the backpropagation algorithm. The researchers had employed a cross-validation approach in the study. The study reveals that feed forward neural network can be successfully used for the diagnosis of thyroid disease. ANN can be used for complex function mapping as it is very much flexible and exhibits a lot of potential in the detection of thyroid diagnosis (Zhang & Berardi, 1998).

Table 1: Summary of the Different Models Used by Researchers

| Study                                                   | Year | Method Used                                                       |
|---------------------------------------------------------|------|-------------------------------------------------------------------|
| Varde AS,<br>Massey KL,<br>Wood HC                      | 1991 | The Cross Validation<br>Variable Selection<br>Regression Method   |
| Sharpe PK,<br>Solberg HE,<br>Rootwelt K,<br>Yearworth M | 1993 | SVM<br>Probabilistic Neural Networks                              |
| Rago T,<br>Chiovato L,<br>Grasso L,                     | 2001 | Multilayer perceptron (MLP), Radial Basis Function Network (RBFN) |

|                                                                                                |      |                                                                                                        |
|------------------------------------------------------------------------------------------------|------|--------------------------------------------------------------------------------------------------------|
| Pinchera A,<br>Vitti P                                                                         |      |                                                                                                        |
| Hoshi K,<br>Kawakami J,<br>Kumagai M,<br>Kasahara S,<br>Nishimura N,<br>Nakamura H,<br>Sato K. | 2005 | RBF, Back Propagation<br>Learning Vector Quantization(LVQ)                                             |
| Paliwal M,<br>Kumar UA                                                                         | 2009 | Learning Vector Quantization (LVQ)                                                                     |
| Saiti F,<br>Naini AA,<br>Shoorehdeli MA,<br>Teshnehlab M                                       | 2009 | Decision Tree Attribute Splitting Rules                                                                |
| Dogantekin E,<br>Dogantekin A,<br>Avci D                                                       | 2011 | MLP With Back Spread<br>Quick Back Propagation (FBP)<br>Conic Segment Capacity Neural Networks (CSFNN) |
| Aziz SB                                                                                        | 2011 | ESTDD (Expert System Thyroid Disease Diagnosis)                                                        |
| Kousarrizi,<br>Nazari MR,<br>Seiti F,<br>Teshnehlab M                                          | 2012 | GDA (Generalized Discriminant Analysis), WSVM (Wavelet Support Vector Machine)                         |
| Bhalla JS,<br>Aggarwal A                                                                       | 2013 | LDA (Linear Discriminant Analysis)                                                                     |
| Rastogi,<br>Bhalla A,<br>Monika A                                                              | 2014 | Fuzzy Cognitive Map (FCM)                                                                              |
| Valanarasi Antony Santiago Vaz                                                                 | 2014 | Expert System Oriented Methodology                                                                     |
| Gireesha HM,<br>Nanda S                                                                        | 2015 | MLP<br>SPSS                                                                                            |
| Pandey S,<br>Gour DK,<br>Sharma V                                                              | 2015 | Multivariate Analysis                                                                                  |
| Prerana PS                                                                                     | 2015 | Levenberg Marquardt Method (LMM)                                                                       |
| Shroff S,<br>Pise S,<br>Chalekar P,<br>Panicker SS                                             | 2015 | AIRS (Artificial Immune Recognition System)                                                            |

A and Babu (2013) proposed SCG Based Feedforward Backpropagation Neural Network (FFBNN) for thyroid classification of normal and abnormal thyroid function. The results of the experiment revealed that SCG Based Feedforward Backpropagation Neural Network (FFBNN) attained better performance when compared to the LMA based FFBNN. Srujana et al. (2016) developed a model for diagnosis of thyroid disease using the artificial neural network. The experiments and results proved that using Feedforward Neural Networks to diagnose disease had an increased impact on the accuracy of prediction. The

proposed model achieved a classification accuracy of 98% for thyroid disease. The feed-forward neural network can be generalized to the other disease diagnosis. (Sibia, Kaur, Singh, Singh, & Neki, 2018) presented a study to find the prevalence of reproductive abnormalities in hypothyroid females and the incidence of other autoimmune diseases in these females. The experiments and results revealed that there is an increased prevalence of reproductive dysfunction in hypothyroid females. Hypothyroidism should be diagnosed early, so that early and adequate replacement of thyroid hormone can prevent the aforementioned complications. Corinne R. Fantz et al. (1999) presented reviews of various cases of thyroid activity that occurs during pregnancy. The study revealed that proper diagnosis and treatment of thyroid dysfunction during pregnancy is important. During the periods of pregnancy, assessment of both hypo and hyperthyroidism is advised along with the careful clinical evaluation of other symptoms as well as measurement of Thyroid Stimulating Hormone (TSH) and free thyroid hormones (C R Fantz, Dagogo-Jack, Ladenson, & Gronowski, 1999). Prema B (2010) presented a study of thyroid screening of eighty-two women during pregnancy. The study revealed that screening of T3, T4, FT4, and TSH is able to pick up even borderline cases 0.25- 2mIU/ml recognized as a normal reference range for TSH. Adequate replacement therapy should be given when TSH is above 3mIU/ml and/or with low T4, FT4 in pregnancy.

### 3. Research Method

The objectives of this study are 1) to prepare a model to predict the identification of thyroid malfunction in pregnant women, 2) compare the accuracy of prediction with traditional prediction tools like random forest and multiple regression. For the purpose of the study, we have used the Thyroid Data Set (Quinlan, 1986; Quinlan, Compton, Horn, & Lazarus, 1986) available in the UC Irvine Machine Learning Repository (Dheeru & Karra Taniskidou, 2017). The data consists of two files; one of them is employed to train the data set which is having 3772 cases without any missing values, the other file consists of the test dataset on which the predicted model is to be applied and the model is expected to give occurrence of thyroid disease as a resultant. The test data set consists of 3428 cases without any missing values. Both the files consists of 21 attributes out of which 15 attributes are binary and rest 6 are continuous.

### 4. Data Preprocessing

Before using the data for the analysis, it must be checked for any errors. Since the data is not having any missing values, hence, no treatment is required for missing values. We have checked for any outliers through Mahalanobis distance (Mahalanobis, 1936) and there were 385 cases having Mahalanobis Distance greater than 47 and the corresponding significant value to be less than 0.001, which was calculated using the Cumulative Chi-Square Distribution Function, leaving only 3387 cases which can be used to build the model. When the results of two different scenarios i.e. training and testing dataset with and without outliers included, were compared, the result exhibited that better prediction results were obtained with the inclusion of outliers in the training and testing dataset as the dataset with outliers could amalgamate and explain more variance as compared to the dataset without outliers.

### 5. Experiment And Results

#### 5.1 Environment

We have used the PC having 4 GB of RAM with Intel i3 Processor having 1.90GHz to run the analysis. We have used R programming to evaluate the efficiency of prediction of thyroid abnormalities in pregnant women and have used Artificial Neural Network (Deep Learning Method). Initially, the Training Dataset was used to train the model and then the accuracy of the model was first tested on the Training Dataset and latter was tested on the Testing Dataset.

#### 5.2 Evaluation Measures

For our experiments, we have used common measures for the evaluation of the classification quality: Accuracy, Precision, Recall, and F-Measure (T. Y. Chen, Kuo, & Merkel, 2006; POWERS, 2011).

Table 2: Confusion Matrix

|        |          | Detected            |                     |
|--------|----------|---------------------|---------------------|
|        |          | Positive            | Negative            |
| Actual | Positive | True Positive (TP)  | False Negative (FN) |
|        | Negative | False Positive (FP) | True Negative (TN)  |

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A ratio of a total number of correctly predicted cases i.e. True Positive and True Negative to all the cases, represent Accuracy. Precision refers to the ratio of True Positive cases to the sum of True Positive and False Positive. Recall refers to the ratio of True Positive to the sum of True Positive and False Negative (Fawcett, 2005). Additionally, we have used F-Measure to combine the recall and precision which is considered a good indicator of the relationship between them (Chen et al., 2006).

In order to remove the deficiency of the Accuracy and Recall, Matthews Correlation Coefficient has been used to better understand the efficiency of the confusion matrix. Matthews Correlation Coefficient proves to be the most informative as compared to Accuracy and F-Measure (F1 Score) in evaluating the binary classifications, as it takes into account True and False Positives and Negatives and is usually considered as a balanced measure which can even be used in the cases when classes of different sizes are used (Chicco, 2017).

### 5.3 Evaluation Results

For the comparison with the basic models, we have first calculated the prediction accuracy of the model with the help of multiple regression method with the help of SPSS. The  $R^2$  value came to be 0.395 and the model summary exhibited the significance of the model at a 5% significance level indicating a very weak relationship between the independent and dependent variable. Based on the coefficients found with the help of multiple regression, we have predicted the values thyroid deficiency and have compared with the actual values to find out the value of Root Mean Square Error (RMSE), representing an absolute measure of fit, which came out to be 0.294. The coefficients derived from the linear regression were used to predict the values of detection of thyroid disease in Train Dataset and in the resultant, the number of correctly predicted cases were 74 (1.96%) including the cases where the error value was ignored up to one decimal place and taken into count as correctly predicted values. The Accuracy, Precision, Recall, and F-Measure values are 1.96%, 0.03%, 0.35%, 0.05% respectively. When the predicted model, obtained from linear regression, was used to predict the values of thyroid deficiency and when compared with the actual values of Test data set, the number of correctly predicted cases were 83 (2.42%) including the cases where the error values were ignored after one decimal places and taken into count as correctly predicted values. Confusion matrix was prepared on the bases of predicted values and actual values of the Test Dataset and resultant Accuracy, Precision, Recall, and F-Measure values were 2.42%, 0.03%, 0.40%, and 0.06% respectively.

An ensemble learning method of classification, Random Forest, functions by creating an assembly of decision trees on the training dataset and generates the output model, which is the mode of classes or mean prediction of individual trees (Ho, 1995). For R programming, a Random Forest Algorithm was developed by (Liaw & Wiener, 2018), which is an extension of the random forest algorithm (Breiman, 2001). Random Forest Algorithm was used for first, developing the model and then predicting the detection of thyroid disease. As the data is in two parts, one the Train Data and the Test Dataset. First, the model was developed and then the accuracy of predicting the output was checked on the Train Dataset itself to check the accuracy of the model and then the model was used to predict the output of test data set. Once the output for Test Dataset was predicted, it was compared against the actual values of the output and based on confusion matrix prepared, the Accuracy, Precision, Recall, and F-Measure values for the test data set are 22.462%, 7.990%, 91.600%, and 14.698% respectively. The RMSE values for both Train data set and test data set are 7.93% and 11.82% respectively. Artificial Neural Network and Deep Learning was used to precisely predict the values of deficiency of thyroid in pregnant women and to assess the accuracy of the prediction confusion matrix was used. The data, when tested, identified the model which can represent the data and the same was employed to predict the output values of deficiency of thyroid, which is a dependent variable. Both the values i.e. actual deficiency of thyroid and the predicted values of deficiency of thyroid were employed to prepare a confusion matrix. The following Table 3 represents the confusion matrix for the training dataset.

Table 3: Confusion Matrix for Train Data Set

|              |   | Predicted class |     |      | Total | Correctly Predicted | Correctly Predicted % age | Error | Error % age |
|--------------|---|-----------------|-----|------|-------|---------------------|---------------------------|-------|-------------|
|              |   | 1               | 2   | 3    |       |                     |                           |       |             |
| Actual class | 1 | 89              | 2   | 2    | 93    | 89                  | 95.70%                    | 4     | 4.30%       |
|              | 2 | 0               | 190 | 1    | 191   | 190                 | 99.48%                    | 1     | 0.52%       |
|              | 3 | 1               | 3   | 3484 | 3488  | 3484                | 99.89%                    | 4     | 0.11%       |
| Grand Total  |   | 90              | 195 | 3487 | 3772  | 3763                | 99.76%                    | 9     | 0.24%       |

Above Table 3 indicates that out of total 3372 cases, the total number of the correctly predicted case are 3763 (99.76%), meaning only 9 (0.24%) cases are incorrectly predicted, which is a very small percentage of cases predicted inaccurately.

Table 4: Confusion Matrix Calculations for Train Data Set

|           |         |
|-----------|---------|
| Accuracy  | 99.761% |
| Precision | 98.587% |
| Recall    | 98.239% |
| F-Measure | 98.413% |

Table 4 reveals that the Sensitivity/True Positive Rate (TPR) is 98.239% and the Precision/Positive Prediction Rate is 98.578%. The Harmonic Mean of Precision and Sensitivity (F-Measure) is 98.413%. Matthews Correlation Coefficient (Matthews, 1975), which represents the measure of the quality of binary classifications is 98.28%. The model, which was trained on training data set, was then applied to the testing data set having a total of 3428 cases and the resultant confusion matrix is shown below in Table 5.

Table 5: Confusion Matrix for Test Data Set

|              |   | Predicted class |     |      | Total | Correctly Predicted | Correctly Predicted % age | Error | Error % age |
|--------------|---|-----------------|-----|------|-------|---------------------|---------------------------|-------|-------------|
|              |   | 1               | 2   | 3    |       |                     |                           |       |             |
| Actual class | 1 | 63              | 9   | 1    | 73    | 63                  | 86.30%                    | 10    | 13.70%      |
|              | 2 | 0               | 166 | 11   | 177   | 166                 | 93.79%                    | 11    | 6.21%       |
|              | 3 | 11              | 29  | 3138 | 3178  | 3138                | 98.74%                    | 40    | 1.26%       |
| Grand Total  |   | 74              | 204 | 3150 | 3428  | 3367                | 98.22%                    | 61    | 1.78%       |

Above table 5 indicates that out of total 3428 cases, the total number of the correctly predicted case are 3367 (98.22%), leaving only 61 (1.78%) cases, which are incorrectly predicted, which is a very small percentage of cases predicted inaccurately.

Table 6: Confusion Matrix Calculations for Test Data Set

|           |         |
|-----------|---------|
| Accuracy  | 98.221% |
| Precision | 85.130% |
| Recall    | 91.600% |
| F-Measure | 88.247% |

The table 6 exhibits that the Sensitivity/True Positive Rate (TPR) is 91.60% and the Precision/Positive Prediction Rate is 85.13%. The Harmonic Mean of Precision and Sensitivity (F-Measure) is 88.247%. Matthews Correlation Coefficient (Matthews 1975), which represents the measure of the quality of binary classifications is 87.354%. To evaluate the results obtained from the application of Deep Learning algorithm both for Actual and Predicted Values of Training and Testing Dataset, they have been compared with the results obtained from the Random Forest Algorithm and Multiple Regression. And to compare the results, we have used Accuracy, Precision, Recall, F-Measure, and Matthews Correlation Coefficient which is shown below in Table 7.

Table 7: Confusion Matrix Results

| Performance Parameters                 | Deep Learning (ANN) |              | Multiple Regression |              | Random Forest |
|----------------------------------------|---------------------|--------------|---------------------|--------------|---------------|
|                                        | Training Data       | Testing Data | Training Data       | Testing Data |               |
| Accuracy                               | 99.76%              | 98.22%       | 1.96%               | 2.42%        | 22.46%        |
| Precision                              | 98.59%              | 85.13%       | 0.03%               | 0.03%        | 7.99%         |
| Recall                                 | 98.24%              | 91.60%       | 0.35%               | 0.40%        | 91.60%        |
| F-Measure                              | 98.41%              | 88.25%       | 0.05%               | 0.06%        | 14.70%        |
| Matthews Correlation Coefficient (MCC) | 98.28%              | 87.35%       | -88.05%             | -85.41%      | 6.06%         |

The above Table 7 clearly indicates that Deep Learning algorithm has better performance as compared to the other methods based on performance parameters.

## 6. DISCUSSION

The chronological research upholds that the development of various computational techniques evolved to assist medical practitioners to diagnose the presence of disease. Various computational techniques have advanced through the very simple techniques and algorithms are transformed into the advanced and sophisticated data analysis techniques which enable to accurately diagnose the disease. The capacity of the modern tools enables the medical practitioners to harness the power of intelligent systems like Artificial Neural Network (Deep Learning) to train large datasets and to deliver more precise prediction for the diagnosis of disease. In addition to that, medical practitioners can employ prevailing machine learning algorithms, which provide tools that can extensively help them to uncover important relationships between various factors and understand the real causes behind disease and help to improvise the overall ability to diagnose the disease more accurately.

### 6.1 Summarization of Findings

The framework primarily serves two impending roles:

1. The ability to Train and Predict the model based on various data sets obtained from various clinical parameters.
2. The basis for Clinical Artificial Intelligence (CAI) that can “Think Like a Doctor (Bennett & Hauser, 2013).”

The aim of the paper was to develop a model that can effectively and accurately predict the thyroid disease by using deep learning framework in an attempt to address the fundamental healthcare challenges. The current framework takes advantage of new age computational approaches to learn from clinical data and non-clinical data and to develop multifaceted models, which not only helps in diagnosis of thyroid disease but helps in building alternative successive decision paths. The current findings reveal the interestingly high ability of the model to predict the thyroid disease as compared to the models like Multiple Regression and Random Forest. The results as shown in Table 7, demonstrates the notable ability of the Deep Learning Model to predict the outcome with greater accuracy and precision.

### 6.2 Future Work

The very limited portion of what CAI can achieve has been tested here, there are many areas suitable for exploration in this domain. Further expansion to what work has been done can be carried out in the fields like:

- **Variable Factors for Doctors:** the decision making of doctors very much depends on the prior experience of the individual doctor.
- **Variable Factors for Patients:** Patients exhibits different behavior i.e. missing appointments, missing medications, not following the treatment fully.
- **Estimate Time To Recover:** Time taken by each and every individual patient to completely recover.
- **Optimal Personalized Models:** Application of machine learning tools to customize optimal treatment and effect of treatment for the individual patient.
- **Individualistic Model:** Data could be compiled, for each individual patient, to understand the transition, which can be made available to the doctor at the time of decision making.
- **Combining Global Data:** Establishing a connection between all data points for comparison of the huge data available globally and locally so as to understand the similarity and differences between symptoms, diagnoses, treatment and effect of treatment.
- **Adding Gene Data:** The CAI can also be tested on Gene data to predict the possibility of occurrence of specific disease and adopting preventive measures for that.
- **Lifestyle Factors:** Lifestyle factors play a major role in many of the diseases which are not gene-based, hence studying Lifestyle factors would significantly improve the ability of future models to improve the power of prediction.

### 6.3 Conclusion

The main advantage of building Deep Learning Model, for prediction, is to predict for each individual, the chances of occurrence of disease and even the possible results of a particular treatment, which could be used to determine the patient-specific optimal action/treatment to be employed now. The enormous issues related to ethical side of the use of CAI has been discussed in the extant literature recently, which even assimilates the questions pertaining to the role

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of doctor and associated remuneration (performance related pay) (Rosenthal, 2008). Few researchers believe that CAI should be allowed to be used as an augmented tool to be used in predicting and let the doctor focus on actual patient care (Institute of Medicine, 2011). This paper adds to the existing body of knowledge that complex process of diagnosis may be better handled through modeling than the subjective evaluation or intuition alone (Bennett & Hauser, 2013; Meehl, 1986; Schaefer, Bailey, Shechter, & Roberts, 2004).

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